

CS671 – Deep Learning and Its Applications

Technical Documentation Report

Model: CycleGAN (Brain)

Project Group: 5 Region: Brain Modality: Unpaired (CT $\leftrightarrow$ MRI)

1. ARCHITECTURAL OVERVIEW

CycleGAN enables unpaired image-to-image translation by learning a mapping $G : X \rightarrow Y$ and $F : Y \rightarrow X$ such that the dynamics of the domains are preserved through cycle consistency. In this implementation for Brain CT-MRI translation, we utilize a ResNet-based generator and a PatchGAN discriminator to ensure high-fidelity anatomical preservation without requiring perfectly aligned training pairs.

1.1 Generator Architecture (ResNet-9)

The generator is responsible for the modality transformation. We employ the ResNet-9 architecture, which is optimized for 256×256 resolution medical slices.

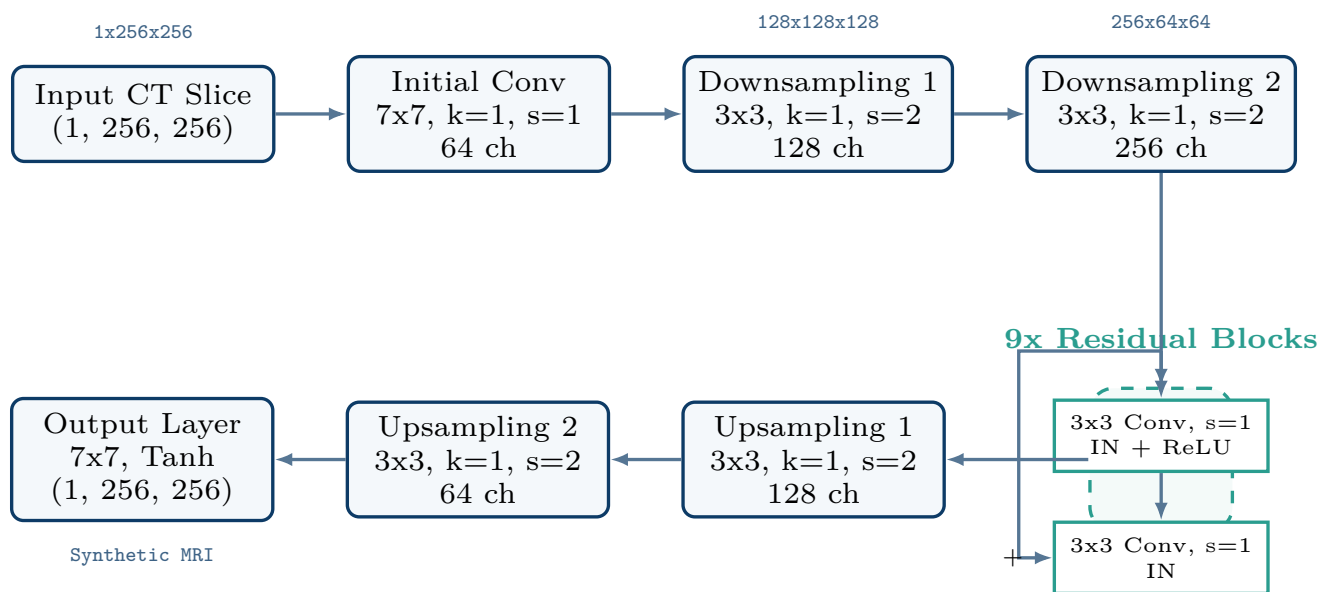


Figure 1: CycleGAN Generator: ResNet-9 architecture with explicit layer-wise components.

Feature Compression (Encoder): The encoder reduces the spatial resolution from 256^2 to 64^2 while expanding the channel depth to 256. This creates a high-dimensional feature representation where modality-invariant anatomical features (like bone density and soft tissue boundaries) are encoded into latent variables.

Deep Residual Learning: The bottleneck consists of 9 stacked residual blocks. Each block computes a residual mapping $H(x) = F(x) + x$. This design is crucial for Brain MRI synthesis because it prevents the "vanishing gradient" problem, allowing the network to learn subtle texture gradients that distinguish grey matter from white matter without losing the global structural alignment provided by the skip connections.

Manifold Reconstruction (Decoder): Two fractional-strided convolutions (transpose convolutions) recover the spatial resolution. The final layer uses a 7×7 convolution with Tanh activation to map the feature space back to the normalized pixel intensity range $[-1, 1]$.

1.2 Discriminator Architecture (70x70 PatchGAN)

The discriminator determines whether a generated MRI slice is realistic compared to real MRI distributions.

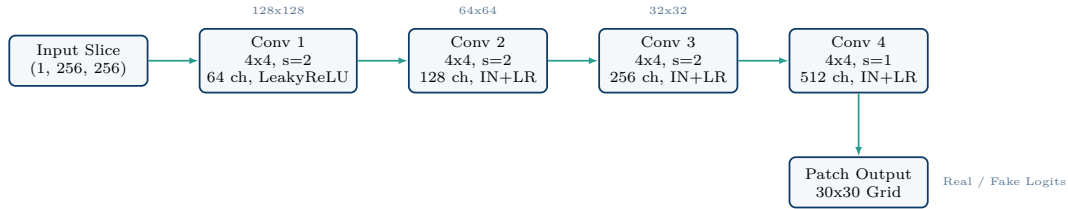


Figure 2: PatchGAN Discriminator: Penalizing structure at the scale of local patches.

Patch-level Evaluation: Instead of outputting a single scalar for the whole image, PatchGAN outputs a 30×30 grid of logits. Each logit corresponds to a 70×70 patch in the original image. This enforces high-frequency consistency (texture) while the cycle-consistency loss handles the global structure (anatomy).

2. TRAINING DYNAMICS

The model is trained using a composite loss function to ensure both realism and cycle consistency.

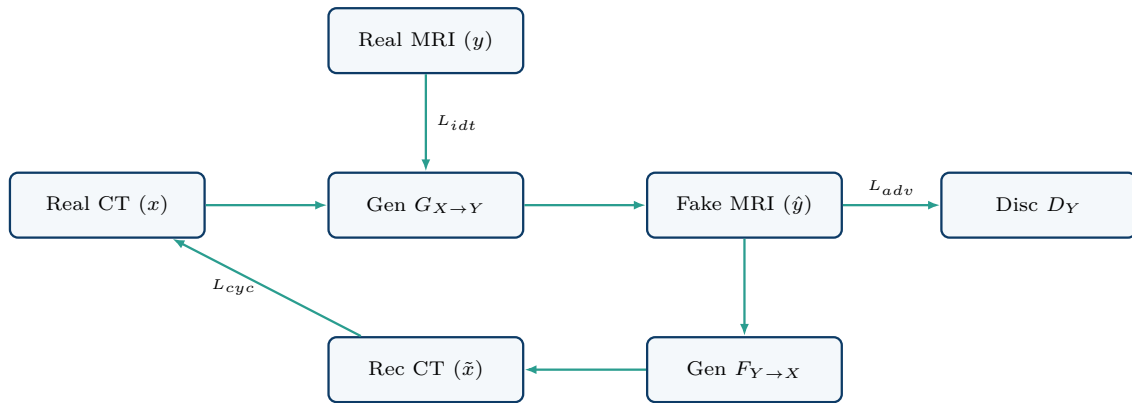


Figure 3: Training Workflow: Interaction between Adversarial, Cycle, and Identity losses.

Loss Objectives:

- **Adversarial Loss (L_{adv}):** Uses LSGAN (Least Squares GAN) for stable training. It forces the generated MRI to follow the distribution of real MRI slices.
- **Cycle Consistency Loss (L_{cyc}):** Enforces $F(G(x)) \approx x$. This is the "soft" supervision that ensures anatomical structures in the CT (skull, brain matter) are correctly mapped to their MRI counterparts.
- **Identity Loss (L_{idt}):** Prevents the generator from changing the color distribution when the target modality is already provided as input, further stabilizing the mapping.

3. INFERENCE PIPELINE

The inference process translates 3D volumes by processing them as a sequence of independent 2D axial slices.

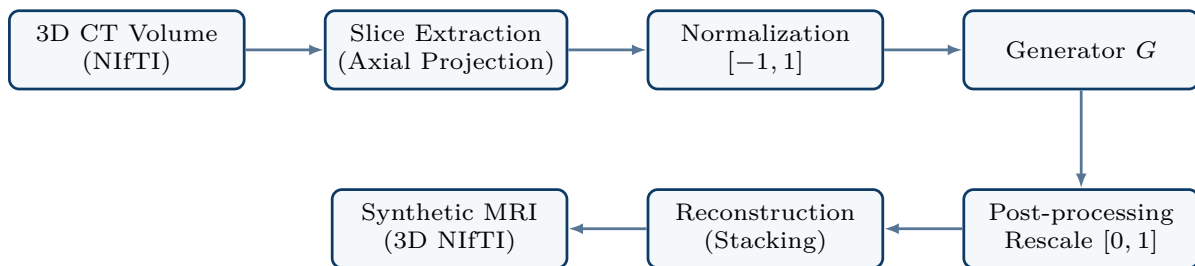


Figure 4: End-to-end inference workflow from volume ingestion to reconstruction.

The inference engine performs intensity normalization to match the training distribution. After the forward pass through the ResNet generator, the synthetic slices are re-stacked into a 3D tensor, ensuring that the affine transformation matrix from the original NIfTI header is preserved for clinical spatial alignment.